Value Functions for Microprice Operators

A Companion Note to
Microprice as a Martingale Corrector

Lev Petersen

Abstract. The companion paper [1] shows that Stoikov's microprice adjustment is the centered solution of a Poisson equation on the post-move chain B, and that the same construction yields martingale correctors for general additive order-book functionals. That development is purely undiscounted: the Poisson equation requires centering because B has eigenvalue 1.
This note, written at the request of colleagues, develops the discounted extension. We introduce a general fixed-point equation $V = u + \Gamma M V$ parametrized by a Markov kernel M and a continuation operator Γ. When $\|\Gamma M\| < 1$, the solution is a convergent Neumann series $(I - \Gamma M)^{-1} u$. We record existence, uniqueness, and norm bounds; catalogue the main special cases (geometric discounting, state-dependent killing, absorbing-chain fundamentals); and show that the undiscounted microprice operator is recovered as the centered limit of discounted resolvents when the discount vanishes.

Keywords: microprice, value function, Neumann series, resolvent, discounted potential, Markov operator, absorbing Markov chain

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Contents

1 Introduction **3**

2 Neumann-series value functions **3**

3 Examples of continuation operators **4**

3.1 Geometric discounting 4

3.2 State-dependent killing 4

3.3 Absorbing-chain fundamentals as resolvents 4

4 Microprice as a centered boundary case **5**

5 Summary **5**

A Proofs **6**

1 Introduction

In [1] we established that Stoikov's microprice [2] is the centered solution of a Poisson equation on the post-move chain. That construction works in an undiscounted setting: centering ($\pi^T h = 0$) is required because the post-move chain B is stochastic and has eigenvalue 1.

Several colleagues have asked how this relates to the discounted value-function machinery familiar from dynamic programming and Markov decision processes [4]. This note answers that question. The main observation is that both the absorbing-chain fundamental matrix $(I - Q)^{-1}$ and the microprice operator $(I - B + \mathbf{1}\pi^T)^{-1}$ belong to a single resolvent family: the general fixed-point equation

$$V = u + \Gamma M V,$$

where M is a Markov kernel and Γ encodes continuation or discounting. When $\|\Gamma M\| < 1$, the inverse $(I - \Gamma M)^{-1}$ exists unconditionally and the solution is a convergent Neumann series. The microprice corresponds to the boundary case $\Gamma = I$, where the resolvent becomes singular on constants and well-posedness is restored by centering.

Notation. We use the notation of [1] throughout. B is the post-move chain (a row-stochastic matrix on a finite state space $\mathcal{X}$ with $|\mathcal{X}| = L$), π its stationary distribution, $\Pi = \mathbf{1}\pi^T$ the projection onto constants, Q the substochastic no-move kernel, T the move kernel ($Q + T$ is row-stochastic), and $N = (I - Q)^{-1}$ the absorbing-chain fundamental matrix. $\|\cdot\|_\infty$ denotes the induced ℓ^∞ operator norm throughout.

2 Neumann-series value functions

Let $M \in \mathbb{R}^{L\times L}$ be a row-stochastic matrix (a Markov kernel on the finite state space $\mathcal{X}$). Let $\Gamma \in \mathbb{R}^{L\times L}$ be a bounded linear operator satisfying the contraction condition

$$\|\Gamma M\|_\infty < 1. \tag{1}$$

Given a bounded payoff $u : \mathcal{X} \to \mathbb{R}$, consider the linear fixed-point equation

$$V = u + \Gamma M V. \tag{2}$$

This template covers a wide range of discounted or randomized-horizon objectives; the only property used below is the contraction condition (1).

Proposition 2.1 (Neumann-series value function)**.** *Assume* (1)*. Then* (2) *has a unique bounded solution given by*

$$V = (I - \Gamma M)^{-1} u = \sum_{k=0}^{\infty} (\Gamma M)^k u. \tag{3}$$

Moreover,

$$\|V\|_\infty \le \frac{1}{1 - \|\Gamma M\|_\infty} \|u\|_\infty.$$

Remark 2.2 (Operator viewpoint). The map $u \mapsto (I - \Gamma M)^{-1} u$ is a bounded linear operator on $\mathbb{R}^L$. When Γ and M are both non-negative, V inherits the sign of u: non-negative payoffs produce non-negative values.

3 Examples of continuation operators

The template (2) separates the *Markov skeleton* M from the *continuation* mechanism Γ. We now record the main special cases that arise in practice.

3.1 Geometric discounting

With $\Gamma = \alpha I$ for $\alpha \in (0,1)$ one has

$$V = u + \alpha MV, \qquad \text{so} \qquad V(x) = \mathbb{E}_x\left[\sum_{n=0}^{\infty} \alpha^n u(X_n)\right].$$

The solution is $V = (I - \alpha M)^{-1}u$, the classical discounted resolvent. The contraction condition is simply $\|\alpha M\|_\infty = \alpha < 1$.

This is the most natural starting point for the microprice connection: as $\alpha \uparrow 1$, the resolvent approaches the undiscounted boundary where centering becomes necessary.

3.2 State-dependent killing

If $\Gamma = D_\gamma := \mathrm{diag}(\gamma(x))$ with $\gamma(x) \in [0,1)$ for each $x \in \mathcal{X}$, then

$$V = u + D_\gamma MV$$

corresponds to a Markov chain that *continues* from state x with probability $\gamma(x)$ and terminates with probability $1 - \gamma(x)$. The contraction condition is $\|D_\gamma M\|_\infty \le \|\gamma\|_\infty < 1$.

This captures objectives stopped at a state-dependent random horizon: from each state, the chain independently decides whether to continue or stop, and the value function accumulates payoffs until termination.

3.3 Absorbing-chain fundamentals as resolvents

Any substochastic kernel Q can be viewed as a killed Markov kernel: at each step the chain either continues (with probabilities given by Q) or is absorbed (with the remaining probability mass). Its fundamental matrix

$$N = (I - Q)^{-1} = \sum_{n \ge 0} Q^n$$

is exactly the value function associated with $\Gamma M = Q$ [3]. The event-time microprice computation in [1] is of this type: Q encodes "continue without a price move," and the absorption condition ($\rho(Q) < 1$) guarantees the Neumann series converges.

In contrast, the post-move chain B is fully stochastic ($B\mathbf{1} = \mathbf{1}$), so $\Gamma M = B$ violates the contraction condition. The undiscounted inverse requires centering, which is the subject of the next section.

4 Microprice as a centered boundary case

For $\alpha \in (0,1)$, the discounted potential $(I-\alpha B)^{-1}$ is well-defined on all of $\mathbb{R}^L$ and admits the Neumann series

$$(I-\alpha B)^{-1}v = \sum_{k=0}^{\infty} \alpha^k B^k v.$$

As $\alpha \uparrow 1$, the resolvent becomes singular because B has eigenvalue 1. But on the centered subspace $\{v : \pi^T v = 0\}$, the limit exists.

Proposition 4.1 (Resolvent limit)**.** *On the centered subspace $\{v : \pi^T v = 0\}$, the discounted resolvent $(I-\alpha B)^{-1}$ converges as $\alpha \uparrow 1$ to the microprice operator $(I - B + \mathbf{1}\pi^T)^{-1}$.*

Remark 4.2 (Behavior off the centered subspace). For a general $v \in \mathbb{R}^L$, decompose $v = (\pi^T v)\mathbf{1} + (v - (\pi^T v)\mathbf{1})$ into its constant and centered parts. Since $B\mathbf{1} = \mathbf{1}$, the resolvent identity

$$(I-\alpha B)^{-1}v = \frac{\pi^T v}{1-\alpha}\,\mathbf{1} \;+\; (I-\alpha B)^{-1}\big(v - (\pi^T v)\mathbf{1}\big)$$

shows that the only obstruction at $\alpha \uparrow 1$ is the constant component, which diverges as $(1-\alpha)^{-1}$. The centered component has a finite limit given by Proposition 4.1.

This clarifies the relationship between the discounted and undiscounted theories: the microprice operator is the "renormalized" limit of the discounted resolvent, obtained by restricting to the centered subspace where the divergent constant mode is absent. In practice, the centering condition $\pi^T G_1 = 0$ (zero stationary drift per move) ensures that the first adjustment already lies in this subspace.

5 Summary

The objects studied in [1] and in this note fit into a single resolvent family:

Absorbing fundamental matrix:	$N = (I-Q)^{-1}$,	$\Gamma M = Q,\ \rho(Q) < 1.$
Discounted potential:	$(I-\alpha B)^{-1}$,	$\Gamma M = \alpha B,\ \alpha < 1.$
State-dependent killing:	$(I - D_\gamma M)^{-1}$,	$\lVert\gamma\rVert_\infty < 1.$
Microprice operator:	$(I - B + \Pi)^{-1}\big\vert_{\ker(\Pi)}$,	centered limit $\alpha \uparrow 1$.

The first three are well-posed Neumann-series inverses under the contraction condition $\lVert\Gamma M\rVert_\infty < 1$. The fourth is the centered boundary case: the resolvent becomes singular on constants, and well-posedness is recovered by restricting to the centered subspace. All share the same algebraic structure; the only difference is whether the continuation operator Γ is strictly contractive or not.

A Proofs

Proof of Proposition 2.1. Since $\|\Gamma M\|_\infty < 1$, the Neumann series $\sum_{k=0}^{\infty}(\Gamma M)^k$ converges absolutely in the induced operator norm, defining a bounded linear operator

$$S := \sum_{k=0}^{\infty}(\Gamma M)^k.$$

Telescoping gives

$$(I - \Gamma M)S = \sum_{k=0}^{\infty}(\Gamma M)^k - \sum_{k=1}^{\infty}(\Gamma M)^k = I,$$

so $S = (I - \Gamma M)^{-1}$. Setting $V = Su$ yields (3), and multiplying by $(I - \Gamma M)$ recovers the fixed-point equation (2).

For uniqueness, if V and W both solve (2), then $V - W = \Gamma M(V - W)$. Taking $\|\cdot\|_\infty$ norms gives $\|V - W\|_\infty \leq \|\Gamma M\|_\infty \|V - W\|_\infty$, hence $V = W$.

Finally,

$$\|V\|_\infty \leq \sum_{k=0}^{\infty} \|\Gamma M\|_\infty^k \, \|u\|_\infty = \frac{1}{1 - \|\Gamma M\|_\infty} \, \|u\|_\infty.$$

□

Proof of Proposition 4.1. Let v satisfy $\pi^T v = 0$ and write $B = \mathbf{1}\pi^T + C$ with $C := B - \mathbf{1}\pi^T$. Since $\pi^T C = 0$ and $C\mathbf{1} = 0$, the centered vector v satisfies $B^k v = C^k v$ for all $k \geq 0$. For $\alpha \in (0, 1)$,

$$(I - \alpha B)^{-1} v = \sum_{k=0}^{\infty} \alpha^k B^k v = \sum_{k=0}^{\infty} \alpha^k C^k v = (I - \alpha C)^{-1} v.$$

By ergodicity, $\rho(C) < 1$, so $(I - \alpha C)^{-1} \to (I - C)^{-1}$ as $\alpha \uparrow 1$. Since $I - C = I - B + \mathbf{1}\pi^T$, the limit is $(I - B + \mathbf{1}\pi^T)^{-1} v$. □

References

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